Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “inertia” to explain your model or result. Sentence starters are optional.

net force: the vector sum of all forces on one object

inertia: the tendency to keep the same velocity without net force

balanced: adding to zero net force in this force context

01 Bring back an earlier idea

A goes 6 m east and 4 m north. B then goes 3 m west and 8 m south.
Draw the route and find the endpoint relative to the start.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Sigma F = 0 \Rightarrow a = 0 \Rightarrow v = \text{constant}$.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

Worked reasoning: A book has 6 N upward support and 6 N downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then sketch two possible velocity-time graphs consistent with this balance.

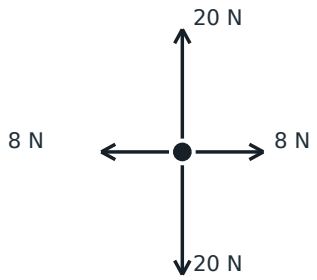
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Balanced while moving

The box is sliding right at steady velocity. Add the horizontal forces shown. Does zero net force mean the box must stop?

Sentence starter: I notice ... . This supports ... because ... .



The dot represents the object. Box moving right

Marzano check - circle: ① ② ③

04 After release

A hand pushes a puck, then loses contact. On an ideal frictionless level surface, what happens to the hand force and the puck’s horizontal motion?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 The bus stops

Explain why a passenger tends to move forward when a bus brakes.
Name what can supply the force that slows the passenger.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A cart accelerates uniformly from rest to 10 m/s east in 3 s.
Find acceleration, average velocity and displacement.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Which motions can have zero net force: resting; straight at steady speed; turning at steady speed; speeding up?
Explain one choice.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A puck slides right at steady speed.
Draw one possible balanced force diagram and explain why a forward force is not required in the frictionless model.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3